Dexamethasone Impact on Covid-19

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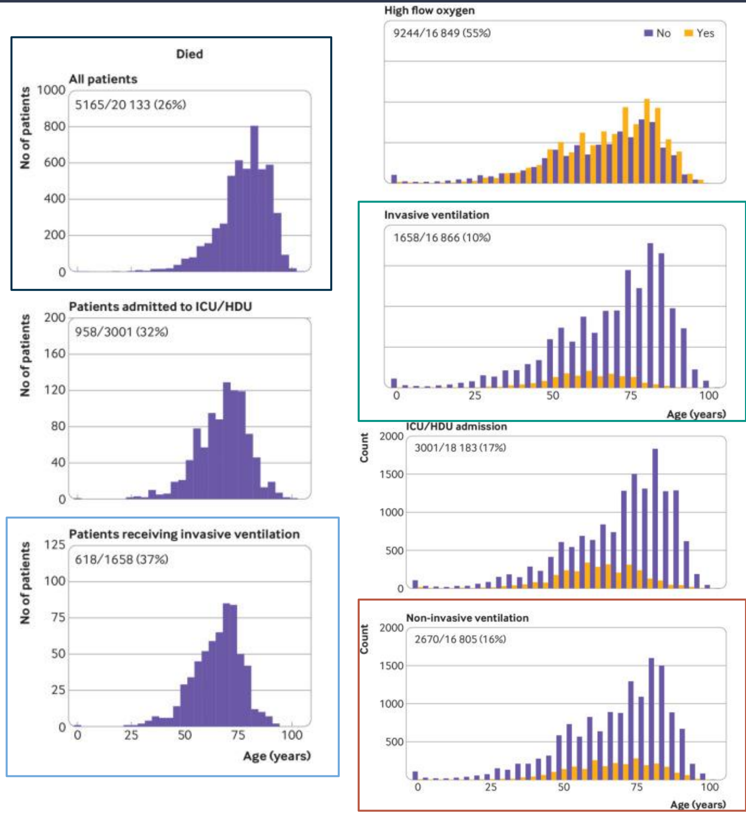
Motivation

- Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) emerged in late 2019 as the cause of Covid-19
- A majority of people were asymptomatic or caught mild symptoms before the disease went away, but a substantial proportion of cases developed respiratory illness requiring ventilation care
- Motivation from a prior trial
  - Design: observational cohort study
  - Sample: 20,133 patients with Covid-19 in the United Kingdom surveyed
  - Results:
    1. 16% required non-invasive ventilation
    2. 10% required invasive ventilation
    3. Among invasive ventilation patients, 37% or 618 patients had passed away, most due to respiratory failure
    4. Overall 26% passed away in the case study
- No therapeutic solutions / treatments at the time could conclusively reduce mortality although treatments such as remdesivir existed that could speed up the recovery process



Motivation (Continued)

- Problem
  - Severe COVID-19 frequently led to acute pneumonia with diffuse alveolar damage and inflammation, similar to other severe viral pneumonias
    - Elevated inflammatory markers (ex: CRP, ferritin, IL-1, IL-6), suggesting inflammation-driven tissue injury
    - This suggests inflammatory immune responses which contribute to organ failure in severe cases
- Solution: Glucocorticoids (like dexamethasone)
  - Glucocorticoid: synthetic steroid hormone used to reduce inflammation and suppress the immune system
  - Few issues
    - Effectiveness and safety were uncertain, specifically for those with covid-19
    - Many guidelines for treatment rarely recommended glucocorticoids with the exception of China for severe cases due to previous studies showing conflicting or inconclusive results
    - Clear need for reliable evidence from a large scale randomized clinical trial to determine true benefit or harm for covid-stricken patients

Design – Type of Study

Feature of study	Traditional RCT	Recovery Trial
Number of Treatments	Tests one intervention/treatment compared to a control or placebo	Multi-arm platform trial with pragmatic design testing several drugs simultaneously against a common control group
Adaptiveness	Fixed design with difficulty inclusion/exclusion of additional treatments	New treatments added overtime with ineffective treatments dropped
Blinding	Randomized, double blinding with rare instances of single or triple blinding to reduce bias	Open-label in order to speed the process of the trial and results, but randomization preserved to avoid confounding results
Scale	Sample size dependent on factors including statistical power, ethics, research design with carefully selected participants	Thousands of subjects chosen for trial with minimal exclusion to reflect real world practice
Speed	Often long in order to see long-term effects and multi-stage process to ensure reliable and safe results	Done quickly through minimal data collection, treatments given as part of normal hospital routines, simple but practical protocol
Objective	To test efficacy of intervention under ideal conditions	Provide a quick assessment and confirmation of readily available treatments for covid-19 in reducing mortality

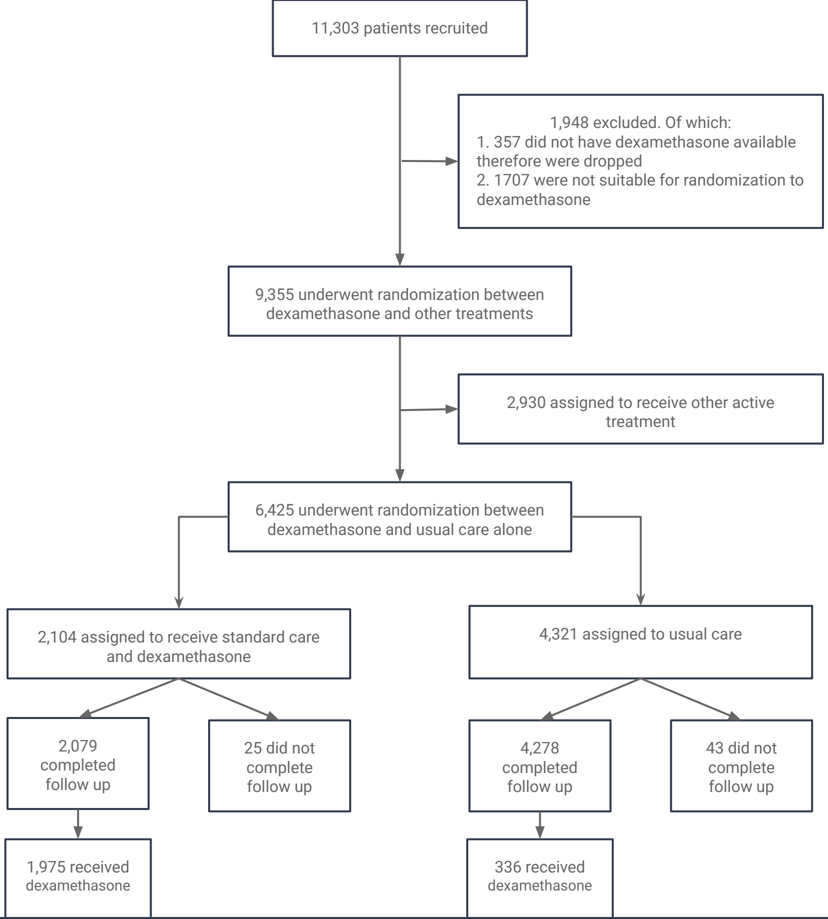
Design – Sample Population

- Sample source: patients hospitalized with Covid-19 at 176 national health service organizations in the United Kingdom
  - Criteria to be a part of the study
    1. Clinically suspected or lab confirmed SARS-CoV-2 infection
    2. No medical history that would put patients at risk if they underwent the trial
    3. Initially 18 years or older but age limit was removed
    4. Written consent from either the patient or legal representative
    5. Hospitalized at baseline of study
  - Other notable features
    - Pregnant or breastfeeding women were eligible to participate
    - Trial was conducted in accordance with the principles of good clinical practice guidelines to ensure the protection of the patients and preserve the integrity of the experiment

Design - Randomization

- Randomization process
  - Tool used: web based randomization system accounting for:
    - Demographic data (Gender, age group, ethnicity)
    - Level of respiratory support
    - Major coexisting illnesses
    - Suitability of treatment for each patient
    - Treatment availability
  - 2:1 ratio of control to treatment sample sizes
    - Control: Standard care
    - Treatment group: Standard care plus a 6 mg dose of dexamethasone once a day for up to 10 days or one of the other treatments being evaluated in the trial
    - Sample sizes of 4000:2000 was determined through wanting at least 90% power at a two sided p-value of 0.01 to detect a clinically relevant proportional reduction of 20% between the 2 groups
- Exceptions
  - Patients were excluded from the study if dexamethasone was unavailable

Due to being an ITT analysis, all 6425 patients were included regardless of completing trial per protocol or not



Randomization Process

Characteristic	Treatment Assignment		Respiratory Support Received at Randomization		
	Dexamethasone (N=2104)	Usual Care (N=4321)	No Receipt of Oxygen (N=1535)	Oxygen Only (N=3883)	Invasive Mechanical Ventilation (N=1007)
Age†					
Mean — yr	66.9±15.4	65.8±15.8	69.4±17.5	66.7±15.3	59.1±11.4
Distribution — no. (%)					
<70 yr	1141 (54)	2505 (58)	659 (43)	2149 (55)	838 (83)
70 to 79 yr	469 (22)	859 (20)	338 (22)	837 (22)	153 (15)
≥80 yr	494 (23)	957 (22)	538 (35)	897 (23)	16 (2)
Sex — no. (%)					
Male	1338 (64)	2749 (64)	891 (58)	2462 (63)	734 (73)
Female‡	766 (36)	1572 (36)	644 (42)	1421 (37)	273 (27)
Race or ethnic group — no. (%)§					
White	1550 (74)	3139 (73)	1221 (80)	2894 (75)	574 (57)
Black, Asian, or minority ethnic group	364 (17)	783 (18)	191 (12)	662 (17)	294 (29)
Unknown	190 (9)	399 (9)	123 (8)	327 (8)	139 (14)
Median no. of days since symptom onset (IQR)¶	8 (5–13)	9 (5–13)	6 (3–10)	9 (5–12)	13 (8–18)
Median no. of days since hospitalization (IQR)	2 (1–5)	2 (1–5)	2 (1–6)	2 (1–4)	5 (3–9)
Respiratory support received — no. (%)					
No oxygen	501 (24)	1034 (24)	1535 (100)	NA	NA
Oxygen only	1279 (61)	2604 (60)	NA	3883 (100)	NA
Invasive mechanical ventilation	324 (15)	683 (16)	NA	NA	1007 (100)
Previous coexisting disease — no. (%)					
Any of the listed conditions	1174 (56)	2417 (56)	911 (59)	2175 (56)	505 (50)
Diabetes	521 (25)	1025 (24)	342 (22)	950 (24)	254 (25)
Heart disease	586 (28)	1171 (27)	519 (34)	1074 (28)	164 (16)
Chronic lung disease	415 (20)	931 (22)	351 (23)	883 (23)	112 (11)
Tuberculosis	6 (<1)	19 (<1)	8 (1)	11 (<1)	6 (1)
HIV infection	12 (1)	20 (<1)	5 (<1)	21 (1)	6 (1)
Severe liver disease‖	37 (2)	82 (2)	32 (2)	72 (2)	15 (1)
Severe kidney impairment**	166 (8)	358 (8)	119 (8)	253 (7)	152 (15)
SARS-CoV-2 test result — no. (%)					
Positive	1865 (89)	3879 (90)	1340 (87)	3433 (88)	971 (96)
Negative	225 (11)	425 (10)	190 (12)	429 (11)	31 (3)
Test result not yet known	14 (1)	17 (<1)	5 (<1)	21 (1)	5 (<1)

Notable facts:

- Mean age 66.1 years old
- On average 7 days since start of covid symptoms
- 36% of patients were female
- History of:
  - Diabetes: 24%
  - Heart disease: 27%
  - Chronic lung disease: 21%
  - At least one major coexisting illness: 56%
  - Lab confirmed Covid-19: 89%
- Patients receiving oxygen during study: 60%
- Patients receiving invasive mechanical ventilation: 16%
- No oxygen support: 24%
- Randomization mostly successful but significant difference in age between dexamethasone and usual care group

Baseline characteristics

Methods

- Outcomes of interest
  - Primary: mortality within 28 days after randomization regardless of cause
  - Secondary
    - Time until hospital discharge
    - Among patients not receiving invasive ventilation at time of randomization, incidence of progression to invasive ventilation or death
  - Other notable interests
    - Cause of death if applicable
    - Occurrence of renal hemodialysis or hemofiltration
    - Occurrence of major cardiac arrhythmia
    - Time and duration of ventilation support received if applicable

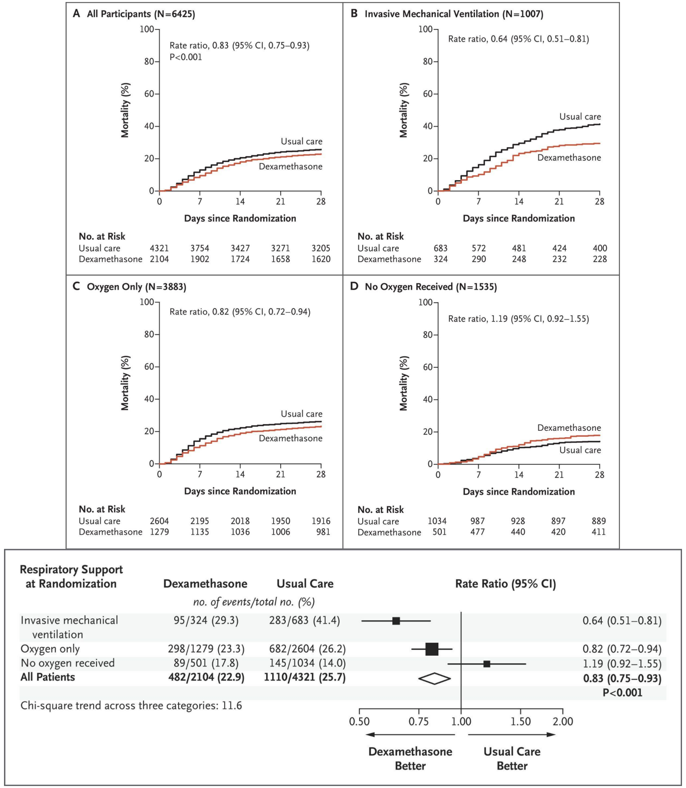
Trial

- Procedure after randomization
  1. All patients received standard care
     a. Patients in the treatment group were given an additional treatment of 6 mg of dexamethasone daily for up to 10 days
     b. Treatment was discontinued if patient was discharged before 10 days were up
  2. Once a patient is either discharged, died, or reached 28 days after randomization they (or a provider) filled out a follow up form which included:
     a. Record regarding patients’ adherence to assigned treatment
     b. Receipt of other trial treatments
     c. Duration of admission
     d. Type and duration of respiratory support
     e. Whether they received renal support
     f. Vital status (including cause of death)
     g. Date of hospital discharge (if applicable)

Statistical Analysis Used

- Hazard ratio
  - Used to estimate the mortality rate ratio for primary outcome of 28 day mortality
  - Handled missing data by assuming patients survived the remainder of the trial duration if follow-up information was incomplete
- Kaplan-meier survival curve
  - Visualize cumulative mortality over the trial period between the dexamethasone and usual care groups
  - Censored data for patients who died during hospitalization before trial period end
- Risk/risk ratio
  - Used to compare the risk of mortality between treatment and the usual care group as well as for subgroups including between those that had invasive mechanical ventilation
- P-values and 95% confidence intervals
  - Used to confirm statistical significance and clinical relevance of treatment effects

Results



All participants

- Dexamethasone significantly reduced 28-day mortality with a 17% risk reduction in mortality

Invasive mechanical ventilation

- Benefitted the most from dexamethasone with 36% risk reduction in mortality as compared to usual care patients
- Clear separation between usual care and dexamethasone curves over time

Oxygen only

- Overall 18% risk reduction in mortality comparing dexamethasone group vs usual care

No oxygen received

- Only group where mortality rates increased with dexamethasone usage vs none
- Conclusions not valid due to insignificant confidence interval

Results – Primary and Secondary Outcomes

Table 2. Primary and Secondary Outcomes and Prespecified Subsidiary Clinical Outcomes.

Outcome	Dexamethasone (N=2104)	Usual Care (N=4321)	Rate or Risk Ratio (95% CI)*
	no./total no. of patients (%)		
Primary outcome			
Death at 28 days	482/2104 (22.9)	1110/4321 (25.7)	0.83 (0.75–0.93)
Secondary outcomes			
Discharged from hospital within 28 days	1416/2104 (67.3)	2748/4321 (63.6)	1.10 (1.03–1.17)
Invasive mechanical ventilation or death†	462/1780 (26.0)	1003/3638 (27.6)	0.93 (0.85–1.01)
Invasive mechanical ventilation	110/1780 (6.2)	298/3638 (8.2)	0.79 (0.64–0.97)
Death	387/1780 (21.7)	827/3638 (22.7)	0.93 (0.84–1.03)
Subsidiary clinical outcomes			
Use of ventilation‡	25/501 (5.0)	65/1034 (6.3)	0.84 (0.54–1.32)
Noninvasive ventilation	20/501 (4.0)	57/1034 (5.5)	0.77 (0.47–1.26)
Invasive mechanical ventilation	9/501 (1.8)	19/1034 (1.8)	1.07 (0.49–2.34)
Successful cessation of invasive mechanical ventilation§	160/324 (49.4)	268/683 (39.2)	1.47 (1.20–1.78)
Renal-replacement therapy¶	89/2034 (4.4)	314/4194 (7.5)	0.61 (0.48–0.76)

Adjustments

- Rate ratios have been adjusted for age

Primary outcome

- Dexamethasone reduced 28 day mortality by 17% compared with usual care

Secondary outcomes

1. Patients receiving dexamethasone were more likely to be discharged alive within 28 days
2. Among patients not on invasive ventilation at baseline, the outcome of progression to invasive ventilation or death during the trial period was reduced by approximately 7% but not statistically significant
3. Dexamethasone reduced mortality by 7% among patients that were not mechanically ventilated at baseline in trial but was not statistically significant

Conclusions

- Overall finding: dexamethasone use for up to 10 days resulted in lower 28 day mortality than usual care
- Subgroup results (after age adjustment)
  - Patients receiving invasive mechanical ventilation: Mortality reduced by 12.3%
  - Patients receiving oxygen only with no invasive ventilation: Mortality reduced by 4.2%
  - Patients not receiving respiratory support: No significant evidence dexamethasone provided a reduction in mortality, and in some cases causing harm rather than help
- Timing of treatment benefit: Greatest benefit for patients who've had covid symptoms for more than 7 days when inflammatory lung damage is likely more pronounced
  - Corresponds to the average window of 1-2 weeks after covid-19 onset when lung damage is most common
- Supporting evidence from prior research: According to a study done in March 2020 for dexamethasone effects on acute respiratory distress syndrome, it concluded that after the 60 day trial period, the mortality rate was 15% lower for those that took dexamethasone compared to usual care which lines up with this trial's 12.3% reduction rate

Conclusions (Continued)

- Dexamethasone was most effective for patients who were more ill, specifically those who needed oxygen or a ventilator, and those who had been sick for more than a week.
- At that stage, the main problem isn't the virus itself anymore but rather the body's overactive immune response which causes inflammation and subsequent damaging of the lungs
- Because dexamethasone suppresses inflammation, it helps calm that immune reaction and prevent further lung injury
- Dexamethasone however is not recommended in the early stages of covid-19 due to the immune system being necessary to fight the virus where the treatment would hinder this task
- These results shouldn't automatically be applied to other viral infections (like flu or MERS), because their timing and disease patterns differ

Strengths of the Trial

- Strengths
    - Randomized controlled trial: allows for causal inference between dexamethasone and reduced mortality of covid-19 patients
    - Real world applicability: since 15% of all hospitalized covid-19 patients from 175+ hospitals in the UK were enrolled, this improved the validity and generalizability of the study's findings
    - Clinically meaningful primary outcome: the 28 day all-cause mortality as the primary outcome of interest is simple yet highly relevant for patient care and clinical decision making
    - Intention to treat analysis: we analyze subjects according to original treatment assignment regardless of compliance or not which minimizes bias and maintains benefits of randomization

Weaknesses of the Trial

- Weaknesses
  - Lack of blinding study: since both patients and clinicians knew the treatment that was being received/given respectively, this could introduce performance and/or decision bias such as difference in care or reporting based on treatment knowledge
  - Short follow up period: since the trial lasted only 28 days, there is limited knowledge on long term outcomes or complications
  - Restricted sample group: the results of this trial cannot be applicable to non-hospitalized or early stage patients with covid-19 since the trial was only tested on hospitalized patients with moderate to severe covid-19
  - The inclusion of non-compliant patients: ITT analysis reduces bias but non-compliant participants could reduce precision and alter observed treatment effect
  - Handling of missing data: the assumption of patients with missing follow-up data survived during the trial phase could lead to an underestimation of the mortality rate

Impact of Study - The Treatment Itself

- Glucocorticoids (including dexamethasone) prior to this study was used in syndromes related to covid-19 including influenza, pneumonia, SARS, and middle east respiratory syndrome
- However, the usage of glucocorticoids for covid-19 treatment was considered not recommended due to insufficient evidence of its benefits and concerns for potential harm
- After the results of the trial were published, dexamethasone was adopted into UK practice the same day to treat hospitalized covid-19 patients
- Guidelines issued by UK medical officers as well as the NIH in the United States updated to include the recommended use of glucocorticoids in patients suffering from Covid-19

Impact of Study (continued)

- The Recovery Trial was the first platform trial to be tested in a pandemic situation
  - Due to dire situation, steps of the experiment were done in quick succession
    - Protocol written in 2 days
    - First patient added to the study in 7 days with the rest added over the next month
    - Results on the efficacy of dexamethasone reported 2 months after the 28 day trial
- Proved that simple, large-scale trials are essential to obtain clear clinical answers
- Possible to reliably answer multiple treatment questions during a pandemic
  - Other treatments including hydroxychloroquine and lopinavir being tested in tandem with dexamethasone were being tested as well as the Recovery Trial moving on to additional treatments including aspirin and convalescent plasma
- Demonstrated the effectiveness of the coordination between the national institute for health research network and UK chief medical officers in uniting hospitals for a nationwide trial to ensure trial success

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